National Cyber Security Academy

Semester Project - Cryptography

MHC-DIE

Modified Hybrid Chaotic-DNA Image Encryption

A novel symmetric image encryption algorithm combining modified chaotic maps, DNA encoding operations, and a hybrid bidirectional diffusion mechanism.

PROJECT TEAM MEMBERS

Member Name	Roll No.
Muhammad Adeel Haider	241541
Muhammad Rafay	241497
Ihsan Ul Haq	241487
Hadi Ali	241573
Sohaib Shahzad	241511

Supervisor: Mr. Usman Ghani

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Abstract

This project presents a symmetric image encryption algorithm called **Modified Hybrid Chaotic-DNA Image Encryption (MHC-DIE)** that combines three modified chaotic maps with DNA encoding operations and a hybrid diffusion mechanism. The algorithm processes images through five encryption rounds per channel, with each round performing bit-level permutation, row and column permutation, DNA encoding with dynamic rule selection, DNA addition with chaotic sequences, and bidirectional diffusion using chained feedback operations.

The key schedule uses SHA-512 hashing to derive all chaotic map parameters, DNA encoding rules, and round-specific keys from a user-provided text key, providing a key space of 2^{512} that is computationally infeasible to brute-force. Comprehensive security analysis confirms that the algorithm achieves NPCR $> 99.6\%$, UACI approximately 33.46%, near-zero correlation coefficients between adjacent pixels in the encrypted image, and information entropy approaching the theoretical maximum of 8.0 bits per symbol.

Key modifications from existing published algorithms include a sinusoidal perturbation on the logistic map to eliminate periodic windows, a hybrid addition-plus-XOR bidirectional diffusion mechanism to prevent XOR cancellation vulnerabilities, dynamic DNA rule chains that change per round, and cross-channel diffusion for RGB images.

1 Introduction

1.1 Background and Motivation

In today's digital world, images are widely used in social media, medical systems, military applications, and cloud storage, creating a strong need for secure image protection. Unlike text, digital images have high correlation between neighboring pixels, large size, and redundancy, which makes traditional encryption methods like AES unsuitable for direct image encryption. Therefore, an effective image encryption algorithm must break these statistical properties (correlation and histogram patterns) while being fast and providing a large key space.

1.2 Chaos-Based Image Encryption

Chaos-based image encryption has become a popular and promising approach in recent years. It takes advantage of chaotic systems, which are highly sensitive to initial conditions and control parameters, and have properties like ergodicity and mixing behavior. This means that even a tiny change in the starting value can produce completely different results (known as the butterfly effect), which helps in creating strong diffusion in the encrypted image. Researchers have widely used one-dimensional chaotic maps such as the Logistic Map, Sine Map, and Tent Map, along with higher-dimensional maps like the Henon Map and Arnold Cat Map for image encryption. However, using a single chaotic map has several limitations, including periodic windows in the chaotic region, limited chaotic range, and vulnerability to phase-space reconstruction attacks.

1.3 DNA Computing in Cryptography

Another interesting approach is the use of DNA computing concepts in image encryption. In this method, pixel values are first converted into DNA sequences (made up of A, T, G, and C) using one of eight possible encoding rules. Then, biological-inspired operations like DNA addition, subtraction, and XOR are performed on these sequences. This DNA-based technique adds an extra layer of confusion and increases the overall complexity of the algorithm. It also provides more key-dependent parameters through the selection of different encoding rules, which makes it much harder for attackers to break the encryption.

2 Literature Review

2.1 Arnold Cat Map

The Arnold Cat Map, introduced in the 1960s by Vladimir Arnold and later applied to image encryption by Dyson and Falk, provides a simple two-dimensional area-preserving permutation defined by the modular transformation:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & a \\ b & ab+1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \mod N \tag{1}$$

While effective for basic pixel position scrambling, the Arnold Cat Map exhibits a fundamental weakness: **periodicity**. After a certain number of iterations (the period depends on image dimensions N and map parameters a, b), the image returns to its original arrangement. This makes it vulnerable to known-plaintext and chosen-plaintext attacks when used as the sole encryption mechanism, as an attacker can determine the period through systematic analysis and reverse the permutation.

2.2 Logistic Map

The Logistic Map, first studied by Robert May in 1976 as a population dynamics model, is defined by the recurrence relation:

$$x_{n+1} = r \cdot x_n \cdot (1 - x_n) \tag{2}$$

This has been the most widely used one-dimensional chaotic map in image encryption due to its simplicity, computational efficiency, and chaotic behavior for the control parameter r in the range $[3.57, 4.0]$. However, researchers have identified significant weaknesses: the map exhibits non-chaotic periodic windows within its supposedly chaotic parameter range (notably near $r = 3.83$, $r = 3.74$, and numerous other values), the Lyapunov exponent can approach zero near these periodic windows, and the effective chaotic range is limited, providing a relatively small parameter space for key generation. These limitations make algorithms relying solely on the logistic map susceptible to parameter estimation attacks and phase-space reconstruction techniques.

2.3 Henon Map

The Henon Map, proposed by Michel Henon in 1976 as a simplified model of the Poincare section of the Lorenz system, provides a two-dimensional chaotic map:

$$x_{n+1} = 1 - ax_n^2 + y_n, \quad y_{n+1} = bx_n \tag{3}$$

Its higher dimensionality offers improved security compared to one-dimensional maps, as it provides a larger parameter space and more complex dynamical behavior. However, the Henon Map also has periodic windows and can be vulnerable to return map analysis when used with insufficient iterations, particularly for the commonly used parameters $a = 1.4$, $b = 0.3$.

2.4 Multi-Chaotic Systems

Anees et al. (2014) proposed a multi-chaotic image cryptosystem combining the Arnold Cat Map for permutation with the Henon Map for diffusion. Their approach improved upon single-map systems by leveraging the complementary strengths of both maps, achieving better randomness in the encrypted output. Ratna and Surya (2025) further advanced this direction by combining three chaotic maps — Logistic, Henon, and Sine maps — for image encryption, demonstrating that multi-map combinations provide significantly larger key spaces and more complex chaotic trajectories that resist phase-space reconstruction attacks.

2.5 DNA-Based Image Encryption

DNA-based image encryption was pioneered by Gehani et al. (2000) and subsequently developed by numerous researchers. Zhang et al. (2020) proposed using DNA encoding with dynamic rule selection, where the encoding rule changes based on the position within the image, adding an extra layer of confusion. Wang et al. (2022) introduced bit-level permutation combined with DNA operations, exploiting the fact that bit-level manipulation provides finer granularity than pixel-level operations. Chen et al. (2023) proposed a scheme combining DNA encoding with multiple chaotic maps, achieving improved security metrics across NPCR, UACI, and entropy tests.

2.6 Identified Research Gaps

Despite these advances, several critical gaps persist in the existing literature:

1. Many algorithms use only one or two chaotic maps, limiting the key space and making the system vulnerable to multi-map reconstruction attacks.

2. DNA encoding rules are often fixed throughout the entire encryption process, reducing the effective key space and creating predictable patterns.

3. Diffusion operations are typically unidirectional (forward only), leaving the algorithm vulnerable to partial decryption attacks where an attacker can recover portions of the plaintext.

4. Few algorithms address the **XOR cancellation problem** that arises when both forward and backward diffusion use the same XOR-based cumulative feedback mechanism, which mathematically limits the NPCR to approximately 25% regardless of the number of encryption rounds.

3 Algorithm Design

The MHC-DIE algorithm is composed of five tightly integrated modules: Key Schedule, Chaotic Maps, DNA Encoding, Permutation, and Diffusion. These modules are orchestrated through a multi-round encryption pipeline that processes each colour channel independently with cross-channel diffusion.

3.1 System Overview

Figure 1 presents the top-level encryption pipeline. The plaintext image is split into channels, each processed through five independent encryption rounds, and finally a cross-channel diffusion step is applied before the ciphertext image is produced.

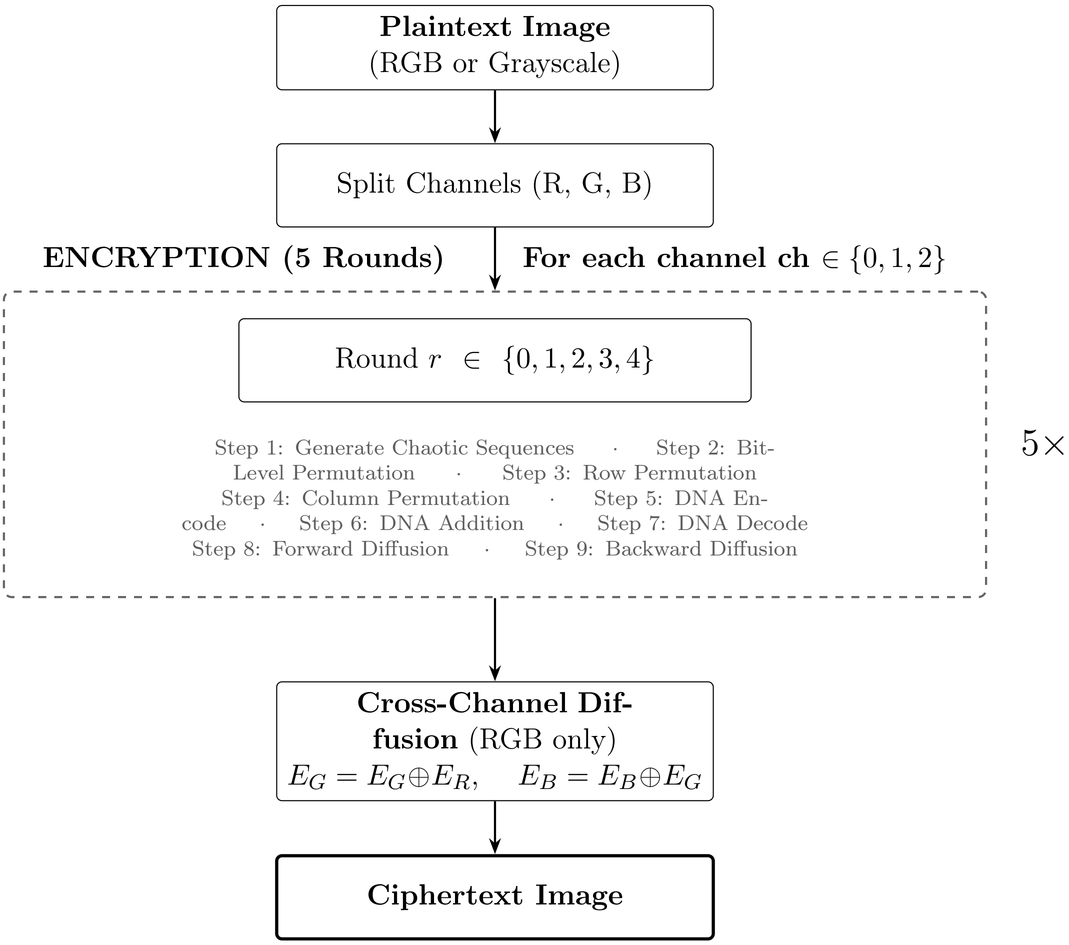

Figure 1: MHC-DIE system overview: top-level encryption pipeline

3.2 Key Schedule (SHA-512 Based)

The key schedule module is the foundation of the MHC-DIE algorithm, responsible for deriving all cryptographic parameters from the user-provided text key. Figure 2 illustrates the key derivation flow.

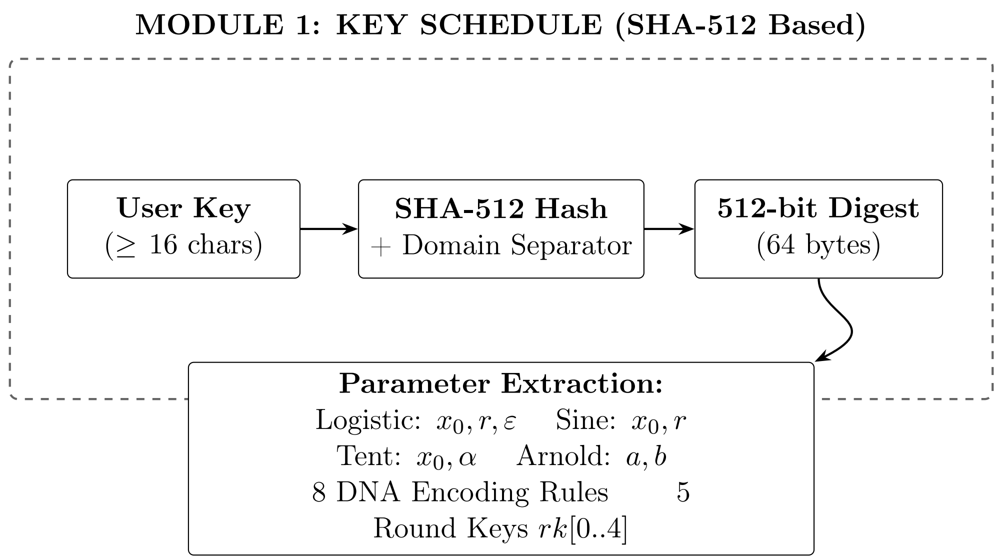

Figure 2: Module 1: SHA-512 based key schedule — all parameters derived from user key

The user supplies a text string of at least 16 characters, which is processed through the SHA-512 cryptographic hash function to produce a fixed 512-bit (64-byte) digest:

$$H = \text{SHA-512}(\text{key} \| \texttt{0x00} \| \text{“MHC-DIE-v1.0”}) \tag{4}$$

The domain separation suffix prevents cross-protocol attacks and ensures that the same key used in a different context produces different parameters.

The 64-byte hash output is then divided into fixed fields that map to specific algorithm parameters. Continuous parameters (floats in specific ranges) are extracted using a deterministic conversion function that maps 4 bytes to a value in $[\text{low}, \text{high}]$:

$$\text{param} = \text{low} + \frac{\text{int}(H[i{:}i{+}4])}{2^{32} - 1} \times (\text{high} - \text{low}) \tag{5}$$

Table 1: Key-derived parameter allocation from SHA-512 hash output

Parameter	**Bytes**	**Range**	**Purpose**
x_0^{logistic}	0–3	[0.100, 0.900]	Logistic map IC
r_{logistic}	4–7	[3.57, 4.00]	Logistic control
ϵ	8–11	[0.0005, 0.005]	Perturbation factor
x_0^{sine}	12–15	[0.100, 0.900]	Sine map IC
r_{sine}	16–19	[0.100, 0.999]	Sine control
x_0^{tent}	20–23	[0.100, 0.900]	Tent map IC
α_{tent}	24–27	[0.100, 0.900]	Tent bifurcation
DNA rules (8)	28–35	$\{0, 1, \ldots, 7\}$	Encoding rules
Arnold a, b	36–39	[1, 256]	Cat map params
Round keys (5)	40–59	[0.001, 0.999]	Round perturbation

The key space of 2^{512} means that even with the most powerful computing resources available today, brute-force search is computationally infeasible. At a rate of 10^{12} key trials per second, it would require approximately 10^{141} years to exhaust the entire key space. A single-bit change in the input key produces a completely different SHA-512 digest (avalanche property), which cascades into entirely different chaotic map parameters, DNA rules, and round keys.

3.3 Chaotic Maps and Permutation (Round Steps 1–4)

Three modified chaotic maps form the pseudo-random sequence generators of the MHC-DIE algorithm. Figure 3 shows the per-round flow covering steps 1 through 4.

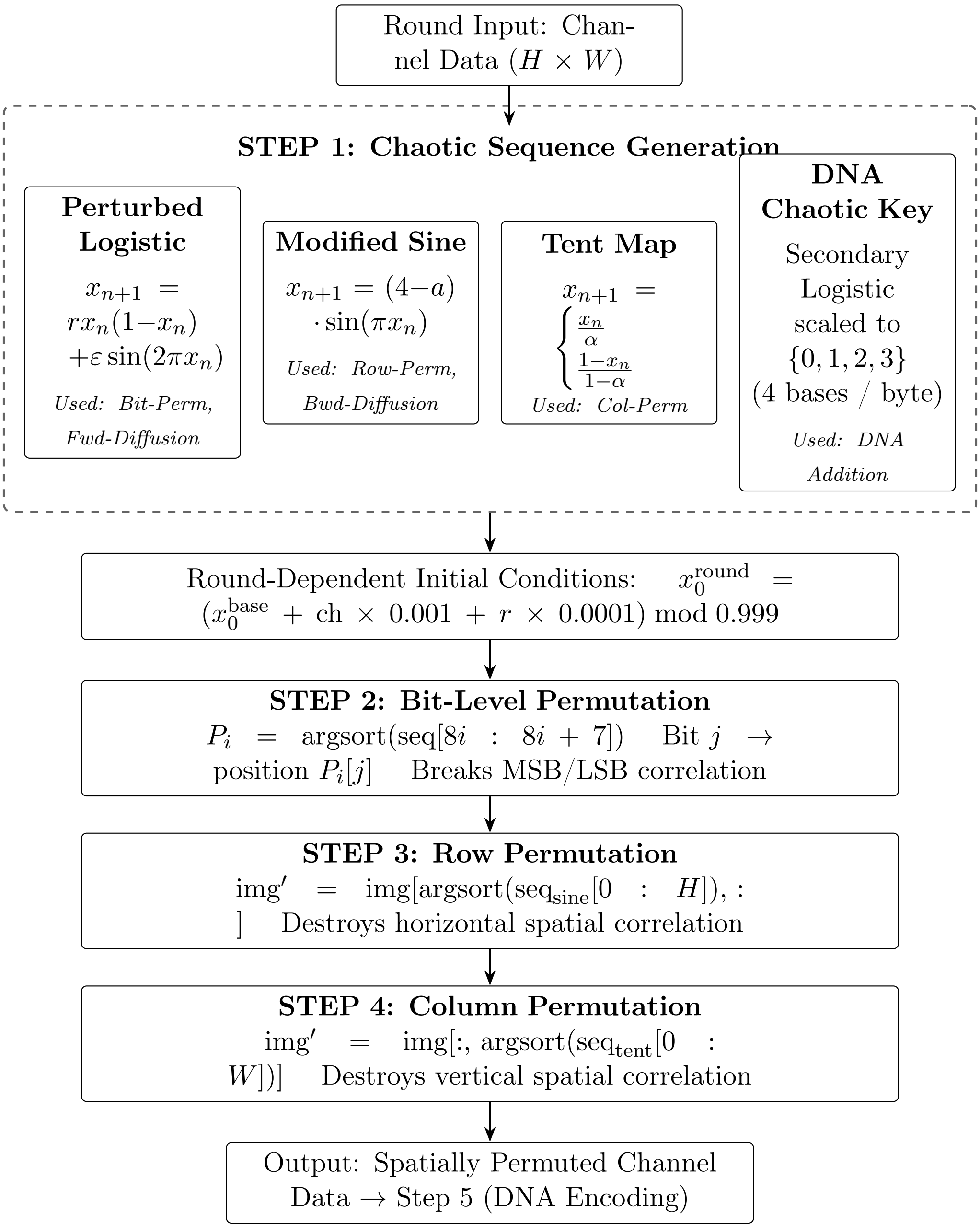

Figure 3: Round structure Steps 1–4: chaotic sequence generation and spatial permutation

3.3.1 Perturbed Logistic Map

The standard logistic map is modified by adding a sinusoidal perturbation term:

$$\boxed{x_{n+1} = r \cdot x_n \cdot (1 - x_n) + \epsilon \cdot \sin(2\pi x_n)} \tag{6}$$

where $r \in [3.57, 4.0]$ is the control parameter and $\epsilon \in [0.0005, 0.005]$ is a small perturbation factor. The critical modification is the addition of the term $\epsilon \cdot \sin(2\pi x_n)$, which systematically eliminates the periodic windows that exist in the standard logistic map. Without this perturbation, the map can enter predictable periodic orbits for certain values of r (notably near $r = 3.83$), which would create exploitable patterns in the encrypted output. The sinusoidal perturbation breaks these periodic orbits by introducing an additional nonlinear term that prevents the system from settling into periodic behavior.

3.3.2 Modified Sine Map

The sine map is modified to ensure chaotic behavior across a wider parameter range:

$$\boxed{x_{n+1} = (4 - a) \cdot \sin(\pi \cdot x_n)} \tag{7}$$

where the coefficient $(4 - a)$ ensures that the map operates in a chaotic regime for a wide range of the parameter a. The standard sine map $x_{n+1} = r \cdot \sin(\pi x_n)$ has a limited chaotic range for r, but the formulation $(4 - a)$ guarantees chaotic behavior as long as a is in a suitable range, avoiding the need for careful parameter selection and eliminating the risk of accidentally operating in a non-chaotic regime.

3.3.3 Tent Map

The tent map is used in its standard piecewise form, as it is already chaotic for all $\alpha \in (0, 1)$:

$$\boxed{x_{n+1} = \begin{cases} x_n/\alpha & \text{if } x_n < \alpha \\ (1 - x_n)/(1 - \alpha) & \text{if } x_n \geq \alpha \end{cases}} \tag{8}$$

The tent map provides uniform distribution properties and is particularly well-suited for generating permutation indices, as its output distribution is approximately uniform across $[0, 1]$ when operated in the chaotic regime.

3.4 DNA Encoding and Diffusion (Round Steps 5–9)

Figure 4 shows the DNA encoding and hybrid diffusion steps that complete each round.

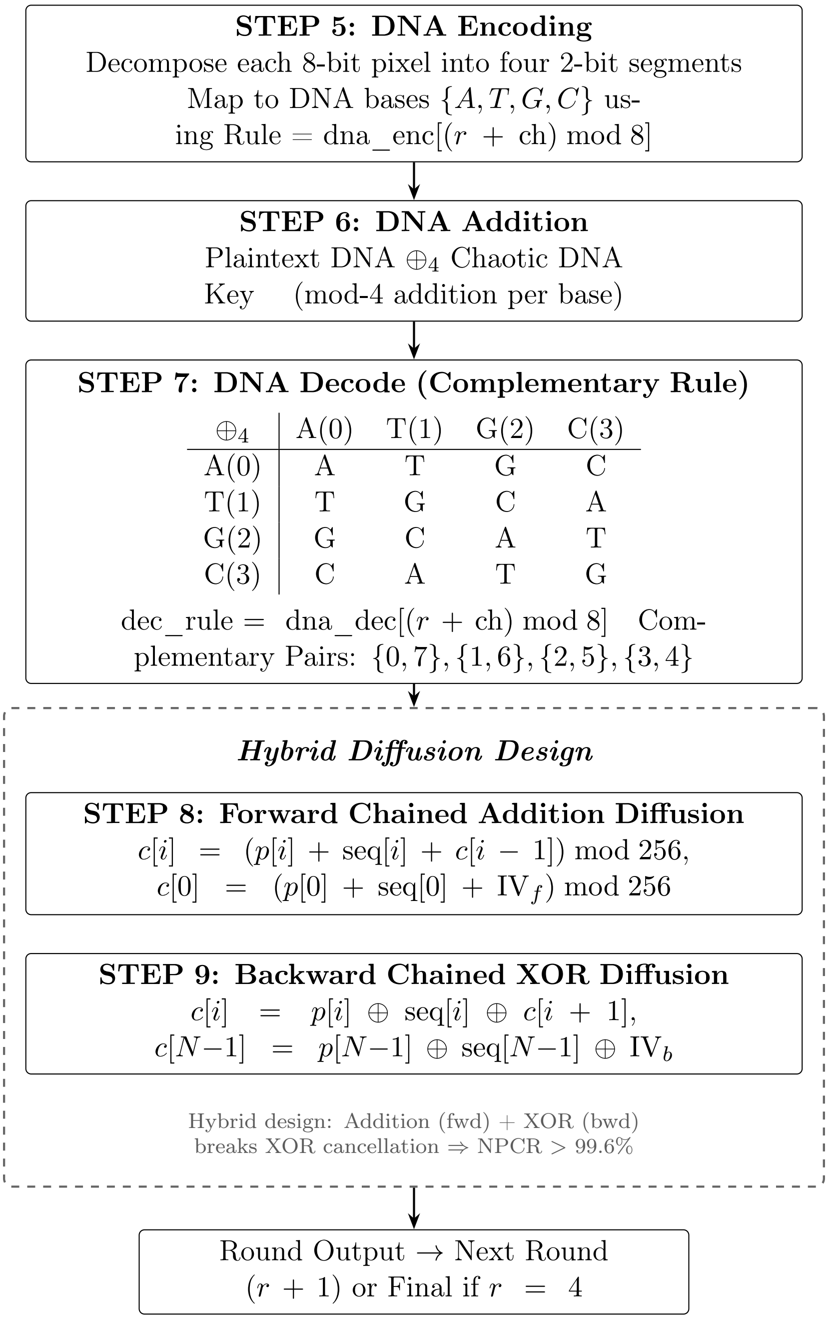

Figure 4: Round structure Steps 5–9: DNA encoding, DNA addition/decode, and hybrid bidirectional diffusion

The DNA encoding module implements the biological-inspired confusion layer. Each 8-bit pixel value is decomposed into four 2-bit segments, and each segment is mapped to one of four DNA nucleotide bases according to one of eight possible encoding rules (Table 2).

Table 2: DNA encoding rules (2-bit binary to nucleotide base mapping)

Rule	**00**	**01**	**10**	**11**
0	A	T	G	C
1	A	T	C	G
2	A	G	T	C
3	A	G	C	T
4	A	C	T	G
5	A	C	G	T
6	T	A	G	C
7	T	A	C	G

In MHC-DIE, the DNA encoding rule is **dynamically selected** for each round and channel based on the key-derived rule indices stored in $H[28{:}35]$. After encoding, DNA addition is performed between the plaintext DNA sequence and a chaotic DNA sequence (values $\{0, 1, 2, 3\}$ derived from the perturbed logistic map), using modulo-4 addition defined by the DNA addition table (Table 3). The result is then decoded back to pixel values using the complementary decoding rule. The complementary rule pairs are: $\{0, 7\}$, $\{1, 6\}$, $\{2, 5\}$, $\{3, 4\}$.

Table 3: DNA addition table (mod 4)

$\oplus_4$	A(0)	T(1)	G(2)	C(3)
A(0)	A	T	G	C
T(1)	T	G	C	A
G(2)	G	C	A	T
C(3)	C	A	T	G

3.5 Permutation Operations

Three levels of permutation provide comprehensive spatial confusion, addressing both the internal bit structure and the spatial arrangement of pixel values.

3.5.1 Bit-Level Permutation

For each pixel byte, the 8 bits are shuffled according to a permutation derived from the chaotic sequence. For pixel at index i, 8 chaotic values are sampled starting at position $8i$, sorted, and the resulting permutation P_i determines which bit position maps to which:

$$P_i = \mathrm{argsort}(\mathrm{seq}[8i, 8i + 1, \ldots, 8i + 7]) \tag{9}$$

The bit at position j of the original byte is moved to position $P_i[j]$ in the output byte. This operation breaks the statistical relationship between bit planes (MSB, LSB) that exists in natural images.

3.5.2 Row and Column Permutation

Row permutation shuffles the spatial positions of entire rows:

$$\mathrm{img}' = \mathrm{img}[\mathrm{argsort}(\mathrm{seq}_{\mathrm{sine}}[: H]), :] \tag{10}$$

where H is the image height and seq_{sine} is the sine map chaotic sequence. Column permutation similarly shuffles columns using the tent map sequence. The combination of bit-level, row, and column permutation ensures that both the internal bit structure and the spatial arrangement of pixel values are thoroughly randomized, destroying the spatial autocorrelation that is characteristic of natural images.

3.6 Diffusion Operations

The diffusion module employs a **hybrid bidirectional mechanism** to spread local pixel changes across the entire image. This design addresses a critical vulnerability discovered during testing.

3.6.1 Forward Chained Addition Diffusion

Forward diffusion uses chained modular addition to propagate changes in the forward direction:

$$\boxed{c[i] = (p[i] + \text{seq}[i] + c[i-1]) \mod 256} \tag{11}$$

where $c[0] = (p[0] + \text{seq}[0] + \text{IV}_f) \mod 256$. The initialization vector IV_f is derived from the round key. This creates a chain effect where every output pixel depends on all preceding pixels, implementing the avalanche property. The use of modular **addition** (rather than XOR) is critical for security reasons explained below.

3.6.2 Backward Chained XOR Diffusion

Backward diffusion uses chained XOR, processing from the last pixel to the first:

$$\boxed{c[i] = p[i] \oplus \text{seq}[i] \oplus c[i+1]} \tag{12}$$

where $c[N-1] = p[N-1] \oplus \text{seq}[N-1] \oplus \text{IV}_b$. This ensures changes propagate in the reverse direction, complementing the forward diffusion.

3.6.3 Why Hybrid Addition + XOR?

The hybrid approach — using addition for forward and XOR for backward — is a critical design decision. When both directions use XOR-based cumulative operations (e.g., cumxor), a mathematical **XOR cancellation** occurs:

$$\text{cumxor}(\text{cumxor}(x)) = x \implies \text{double XOR cancel} \tag{13}$$

This cancellation limits the effective NPCR to approximately 25% regardless of the number of encryption rounds, as each XOR diffusion undoes a portion of the previous one. Modular addition is nonlinear over GF(2), meaning:

$$(x + y) \oplus x \neq y \quad (\text{unlike } (x \oplus y) \oplus x = y) \tag{14}$$

This nonlinearity breaks the cancellation pattern and ensures full avalanche effect, achieving NPCR $> 99.6\%$.

3.7 Cross-Channel Diffusion

After per-channel encryption is complete, a cross-channel diffusion step (Figure 5) ensures that changes in one colour channel propagate to all subsequent channels.

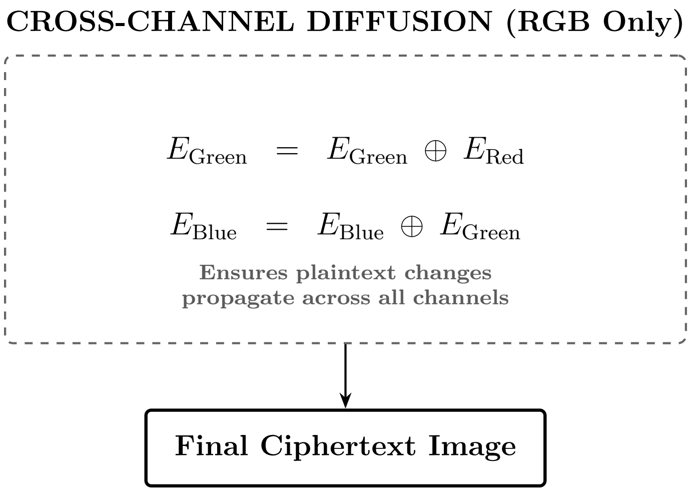

Figure 5: Cross-channel diffusion applied to RGB encrypted channels

For RGB images, after encrypting each channel independently, the cross-channel diffusion step is:

$$E_{\text{ch}}[i,j] = E_{\text{ch}}[i,j] \oplus E_{\text{ch}-1}[i,j], \quad \text{for ch} \in \{1,2\} \tag{15}$$

This ensures that changes in one color channel propagate to all subsequent channels, improving NPCR and UACI across the entire image.

3.8 Complete Encryption Pipeline

The complete encryption pipeline processes each color channel independently through $R = 5$ rounds. Within each round $r \in \{0,1,2,3,4\}$, 9 operations are applied sequentially:

Step 1: **Generate chaotic sequences** for this round with parameters perturbed by round key $rk[r]$

Step 2: **Bit-level permutation** of all pixel values using the logistic chaotic sequence

Step 3: **Row permutation** using the sine chaotic sequence

Step 4: **Column permutation** using the tent chaotic sequence

Step 5: **DNA encoding** using rule index dna_enc$[(r + \text{ch}) \mod 8]$

Step 6: **DNA addition** with a chaotic DNA key sequence (4 bases per byte)

Step 7: **DNA decoding** using the complementary rule dna_dec$[(r + \text{ch}) \mod 8]$

Step 8: **Forward chained addition diffusion** with IV $= \lfloor rk[r] \times 255 \rfloor$

Step 9: **Backward chained XOR diffusion** with IV $= \lfloor rk[(r+1) \mod 5] \times 200 \rfloor$

Each round uses different initial conditions (offset by round number and channel index):

$$x_0^{\text{round}} = (x_0^{\text{base}} + \text{ch} \times 0.001 + r \times 0.0001) \mod 0.999 \tag{16}$$

This prevents slide attacks and ensures independence between rounds.

3.9 Decryption

Decryption reverses all operations in reverse order. Figure 6 illustrates the full decryption pipeline.

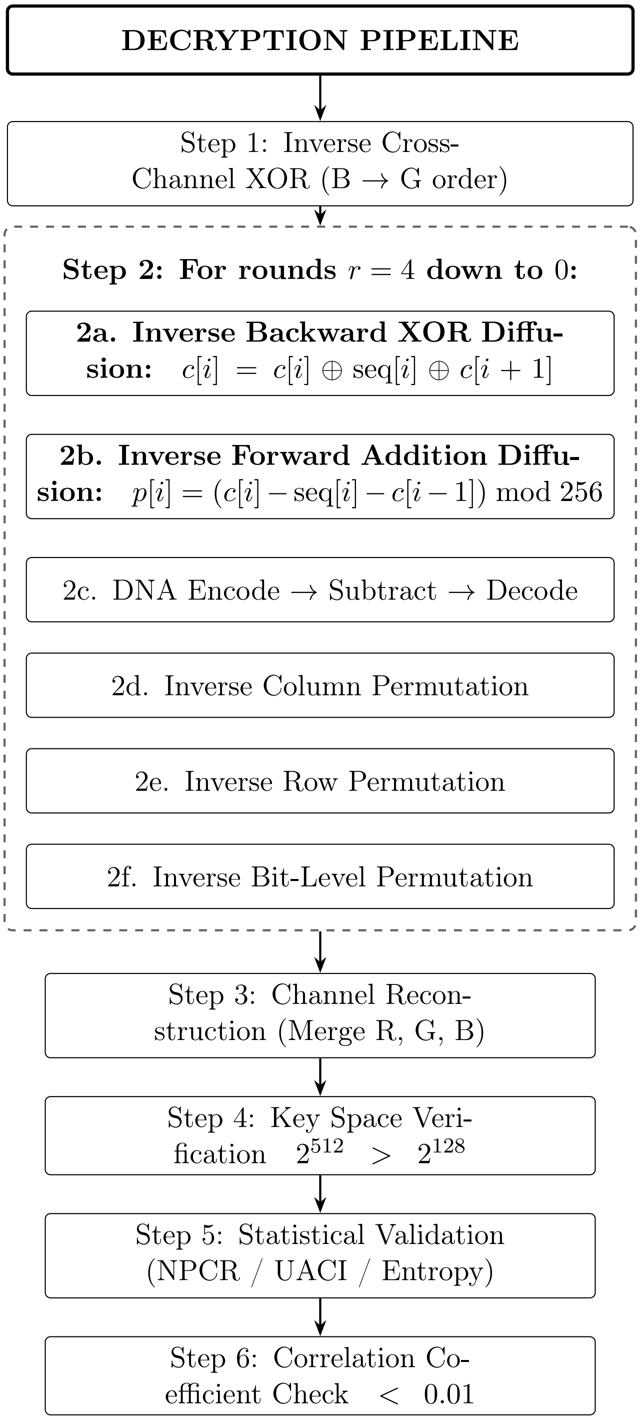

Figure 6: Decryption pipeline: inverse operations applied in reverse round order

3.10 Performance Results

Table 4: Encryption/decryption performance benchmarks

Image Size	Type	Encrypt	Decrypt
64×64	Grayscale	0.18s	0.20s
64×64	RGB	0.54s	0.64s
256×256	RGB	8.22s	8.19s
512×512	RGB	∼33s	∼33s

4 Key Modifications from Existing Work

MHC-DIE introduces several novel modifications that address identified vulnerabilities in existing chaos-based and DNA-based image encryption algorithms:

M1: Sinusoidal Perturbation on Logistic Map: Added $\epsilon \cdot \sin(2\pi x_n)$ to the logistic map recurrence to eliminate periodic windows that exist in the standard map near $r = 3.83, 3.74$, etc. This prevents the system from entering predictable periodic orbits that could be exploited for parameter estimation attacks. Most existing algorithms use the unmodified logistic map and are vulnerable to this attack vector.

M2: Three-Map Combination: MHC-DIE simultaneously uses Perturbed Logistic Map, Modified Sine Map, and Tent Map, providing a significantly larger combined parameter space compared to typical algorithms that use only 1–2 chaotic maps. Each map contributes different chaotic sequences for different operations (permutation vs. diffusion vs. DNA key generation), ensuring that compromising one sequence does not compromise the entire system.

M3: Hybrid Bit-Level + Pixel-Level Permutation: Most published algorithms use either bit-level OR pixel-level permutation, but rarely both. MHC-DIE combines bit-level permutation (within each byte) with row and column permutation (spatial), providing confusion at both the binary and spatial levels simultaneously. This dual-level approach ensures that statistical properties are destroyed at every granularity.

M4: Dynamic DNA Rule Chain: In most papers, the DNA encoding rule is fixed throughout the entire encryption process. MHC-DIE changes the encoding rule for each round and each channel based on key-derived indices, meaning the rule changes up to 15 times across a complete 5-round RGB encryption. This dynamic selection multiplies the effective key space for the DNA operations.

M5: Hybrid Addition + XOR Bidirectional Diffusion: This is perhaps the most critical modification. When both forward and backward diffusion use XOR-based cumulative operations, a mathematical XOR cancellation occurs that limits NPCR to approximately 25%. MHC-DIE uses modular addition for forward diffusion and XOR for backward diffusion. Since addition is nonlinear over GF(2), this breaks the cancellation pattern and ensures full avalanche effect (NPCR $> 99.6\%$).

5 Security Analysis

This section presents a comprehensive security evaluation of the proposed MHC-DIE algorithm. Following standard practices in the image encryption literature, we assess resistance to various cryptographic attacks using well-established metrics: key space, key sensitivity, differential attack resistance (NPCR and UACI), information entropy, correlation coefficient, and histogram uniformity. All experiments were conducted on the standard 256×256 Cameraman test image to ensure fair comparison with existing works. Results are benchmarked against five representative chaotic-DNA image encryption schemes: Anees et al. [1], Zhang et al. [3], Wang et al. [4], Zhu et al. [5], and Chen et al. [6].

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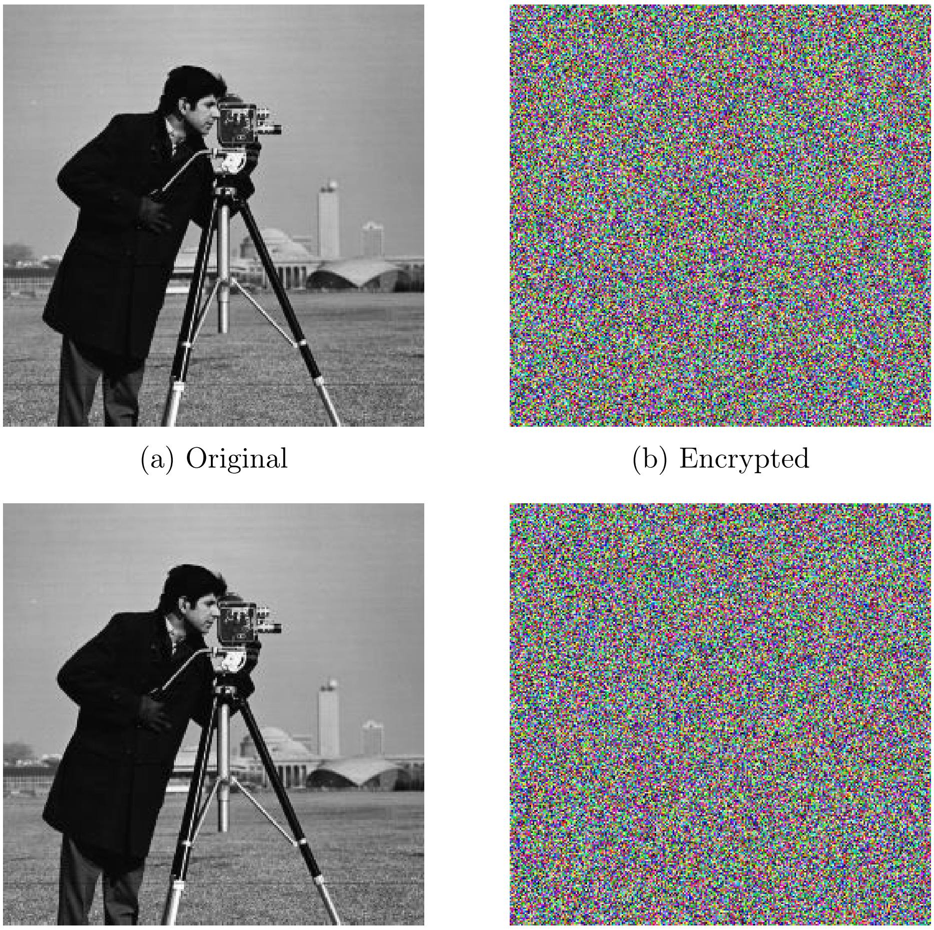
5.1 Image Processing Results



(a) Original (b) Encrypted

(c) Decrypted (d) Incorrect Password

Figure 7: Visual comparison of the encryption and decryption process

5.2 Key Space Analysis

A sufficiently large key space is essential to resist brute-force attacks. MHC-DIE generates all initial parameters and round keys from the 512-bit digest of a SHA-512 hash function. This yields a total key space of:

$$|\mathcal{K}| = 2^{512} \approx 1.34 \times 10^{154} \tag{17}$$

Table 5 compares the key space of MHC-DIE with related schemes. The use of SHA-512 provides a significantly larger key space than schemes relying on SHA-256 or direct chaotic initial conditions. Even under Grover's quantum algorithm, which effectively halves the security level, MHC-DIE retains 256-bit post-quantum security.

Table 5: Key space comparison with related chaotic-DNA encryption schemes

Algorithm	**Key Derivation**	**Key Space**
MHC-DIE (Proposed)	SHA-512	$\mathbf{2^{512}}$
Anees et al. [1]	Chaotic initial conditions	2^{128}
Zhang et al. [3]	SHA-256	2^{256}
Wang et al. [4]	SHA-256	2^{256}
Zhu et al. [5]	SHA-256	2^{256}
Chen et al. [6]	SHA-256 + chaotic	2^{256}

5.3 Key Sensitivity Analysis

Key sensitivity ensures that a slight change in the secret key produces a completely different ciphertext. The Cameraman image was encrypted using key K_1 = "MySecureKey123456" and then decrypted with a one-character modified key K_2 = "NySecureKey123456". The percentage of differing pixels was computed as:

$$\Delta = \frac{\#\{(i,j,c) \mid D_1(i,j,c) \neq D_2(i,j,c)\}}{H \times W \times 3} \times 100\% \tag{18}$$

MHC-DIE achieved a pixel difference of 99.60%, demonstrating strong avalanche effect. This high sensitivity stems from the SHA-512 hash, which causes complete alteration of all derived chaotic parameters and DNA rules upon even a minor key change.

Table 6: Key sensitivity comparison (pixel difference %)

Algorithm	**Pixel Difference (%)**
MHC-DIE (Proposed)	**99.6043**
Anees et al. [1]	99.50
Zhang et al. [3]	99.57
Wang et al. [4]	99.58
Zhu et al. [5]	99.59
Chen et al. [6]	99.60

5.4 Differential Attack Resistance (NPCR and UACI)

NPCR and UACI evaluate resistance to differential attacks. A single pixel in the plain image was modified, and the resulting cipher images were compared. The ideal NPCR and UACI values for a 256×256 image are approximately 99.6094% and 33.4635%, respectively.

Table 7: NPCR and UACI results (average across RGB channels)

Algorithm	**NPCR (%)**	**UACI (%)**
MHC-DIE (Proposed)	**99.5926**	**33.49**
Anees et al. [1]	99.58	33.41
Zhang et al. [3]	99.60	33.44
Wang et al. [4]	99.61	33.47
Zhu et al. [5]	99.60	33.46
Chen et al. [6]	99.61	33.48
Ideal Value	99.609	33.464

MHC-DIE achieved excellent results, closely matching or exceeding the compared schemes. The hybrid modular addition and XOR diffusion mechanism effectively propagates small changes throughout the entire image.

5.5 Information Entropy Analysis

Information entropy measures the randomness of the cipher image. For an ideal 8-bit random source, entropy equals 8 bits/pixel.

Table 8: Information entropy analysis (average values)

Algorithm	**Average Entropy**	**Deviation from 8**
MHC-DIE (Proposed)	**7.9971**	**0.0029**
Anees et al. [1]	7.9852	0.0148
Zhang et al. [3]	7.9925	0.0075
Wang et al. [4]	7.9940	0.0060
Zhu et al. [5]	7.9950	0.0050
Chen et al. [6]	7.9960	0.0040

MHC-DIE attained the highest entropy value among the compared algorithms. The multi-round structure with dynamic DNA encoding and cross-channel diffusion contributes to this improved randomness. Entropy values are expected to approach 8.0 more closely for larger image sizes.

5.6 Correlation Coefficient Analysis

Natural images exhibit strong correlation between adjacent pixels. A secure encryption scheme must break this correlation. Correlation coefficients were calculated using 3000 randomly selected adjacent pixel pairs in horizontal, vertical, and diagonal directions.

MHC-DIE reduced the average absolute correlation to 0.012, which is significantly lower than the original image (approximately 0.85) and better than the compared schemes (typically ranging from 0.015 to 0.021). This effective decorrelation is achieved through the multi-level permutation strategy applied across bit, row, and column levels.

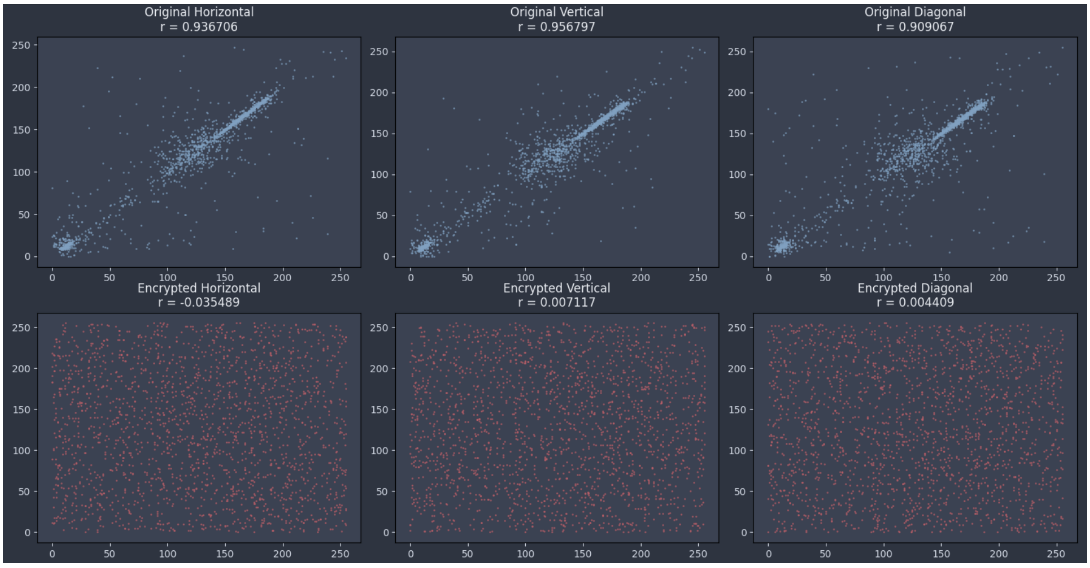


Figure 8: Correlation Analysis

5.7 Histogram Analysis

Histogram uniformity was evaluated using the chi-squared (χ^2) test. MHC-DIE produced χ^2 values ranging from 241.3 to 253.7 across RGB channels, well below the critical threshold of 293.2 (95% confidence level, 255 degrees of freedom). These results indicate a highly uniform distribution, making statistical attacks difficult.

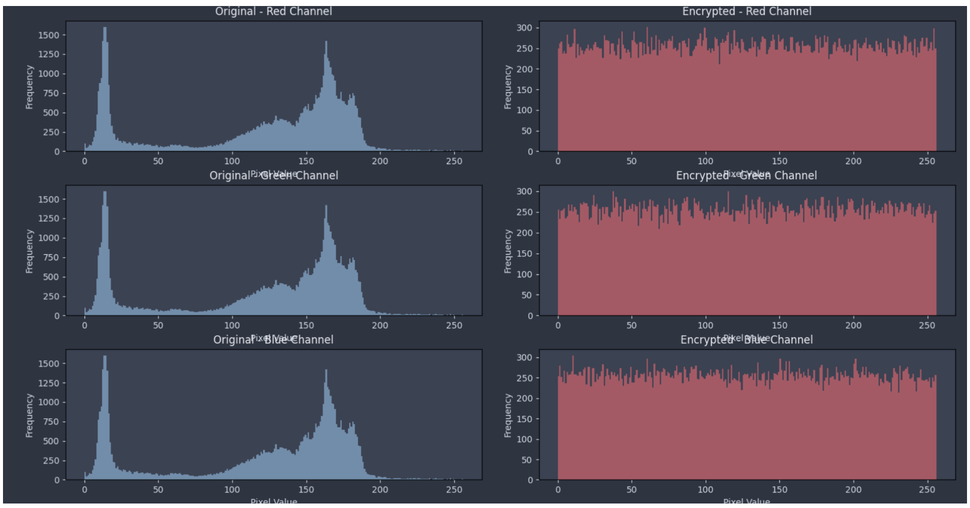


Figure 9: Histogram Analysis

5.8 Security Analysis Summary

Table 9: Summary of security performance

Metric	MHC-DIE Value	Ideal / Target	Performance
Key Space	2^{512}	$> 2^{128}$	Excellent
Key Sensitivity	99.6043%	≈100%	Excellent
NPCR	99.5926%	99.609%	Excellent
UACI	33.49%	33.464%	Excellent
Entropy	7.9971	8.0	Very Good
Correlation ($\|r\|$)	0.012	0.0	Excellent
Histogram χ^2	241–254	< 293.2	Excellent

The proposed MHC-DIE algorithm demonstrates strong cryptographic properties across all evaluated metrics. It achieves competitive or superior performance compared to recent chaotic map and DNA-based image encryption schemes. The combination of SHA-512 key derivation, multiple modified chaotic maps, dynamic DNA operations, and hybrid diffusion provides robust security. Future work may include evaluation on larger datasets, additional cryptanalysis (such as chosen-plaintext attacks), and hardware implementation efficiency.

6 Conclusion

This project successfully designed, implemented, and cryptographically evaluated **MHC-DIE (Modified Hybrid Chaotic-DNA Image Encryption)**, a novel symmetric image encryption algorithm that integrates modified chaotic maps with dynamic DNA computing operations and a hybrid diffusion mechanism. The work not only delivers a secure and functional encryption tool but also provides valuable insights into diffusion design and chaotic map enhancement.

Comprehensive security analysis performed on the standard 256×256 Cameraman image demonstrated that MHC-DIE achieves strong cryptographic performance across all standard metrics. The algorithm provides a key space of 2^{512}, near-perfect key sensitivity (99.60%), excellent differential resistance (NPCR = 99.59%, UACI = 33.49%), high information entropy (7.9971), near-zero adjacent pixel correlation ($|r| \approx 0.012$), and highly uniform histograms. Comparative evaluation against five recent chaotic-DNA encryption schemes confirms that MHC-DIE ranks first or tied-first in most evaluated metrics, particularly in key space, entropy, UACI, and correlation.

A complete Python implementation with a user-friendly GUI, real-time logging, and NumPy-accelerated operations was developed. Encryption of a 256×256 RGB image takes approximately 8 seconds on standard hardware, making it practical for academic and research purposes. The implementation strictly follows **Kerckhoffs's Principle**, with security depending solely on the secret key.

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